R Lab 11. Multicollinearity and Ridge Regression

1. Measuring Multicollinearity

```
> attach(mtcars)
> names(mtcars)
 [1] "mpg"  "cyl"  "disp" "hp"   "drat" "wt"   "qsec" "vs"   "am"   "gear" "carb"
```

 # R shows rather high correlations among variables

```
> cor(mtcars)
```

```
            mpg        cyl       disp         hp        drat         wt        qsec         vs          am       gear        carb
mpg   1.0000000 -0.8521620 -0.8475514 -0.7761684  0.68117191 -0.8676594  0.41868403  0.6640389  0.59983243  0.4802848 -0.55092507
cyl  -0.8521620  1.0000000  0.9020329  0.8324475 -0.69993811  0.7824958 -0.59124207 -0.8108118 -0.52260705 -0.4926866  0.52698829
disp -0.8475514  0.9020329  1.0000000  0.7909486 -0.71021393  0.8879799 -0.43369788 -0.7104159 -0.59122704 -0.5555692  0.39497686
hp   -0.7761684  0.8324475  0.7909486  1.0000000 -0.44875912  0.6587479 -0.70822339 -0.7230967 -0.24320426 -0.1257043  0.74981247
drat  0.6811719 -0.6999381 -0.7102139 -0.4487591  1.00000000 -0.7124406  0.09120476  0.4402785  0.71271113  0.6996101 -0.09078980
wt   -0.8676594  0.7824958  0.8879799  0.6587479 -0.71244065  1.0000000 -0.17471588 -0.5549157 -0.69249526 -0.5832870  0.42760594
qsec  0.4186840 -0.5912421 -0.4336979 -0.7082234  0.09120476 -0.1747159  1.00000000  0.7445354 -0.22986086 -0.2126822 -0.65624923
vs    0.6640389 -0.8108118 -0.7104159 -0.7230967  0.44027846 -0.5549157  0.74453544  1.0000000  0.16834512  0.2060233 -0.56960714
am    0.5998324 -0.5226070 -0.5912270 -0.2432043  0.71271113 -0.6924953 -0.22986086  0.1683451  1.00000000  0.7940588  0.05753435
gear  0.4802848 -0.4926866 -0.5555692 -0.1257043  0.69961013 -0.5832870 -0.21268223  0.2060233  0.79405876  1.0000000  0.27407284
carb -0.5509251  0.5269883  0.3949769  0.7498125 -0.09078980  0.4276059 -0.65624923 -0.5696071  0.05753435  0.2740728  1.00000000
```

 # VIF = variance inflation factors are used to measure multicollinearity.

```
> install.packages("car")
> library(car)

> reg = lm( mpg ~ ., data=mtcars )
> vif(reg)
       cyl      disp        hp      drat        wt      qsec        vs        am      gear      carb
15.373833 21.620241  9.832037  3.374620 15.164887  7.527958  4.965873  4.648487  5.357452  7.908747
```

 # In presence of multiple categories and dummy variables, generalized VIF are used, adjusted for the degrees of freedom (number of dummies per a categorical variable)

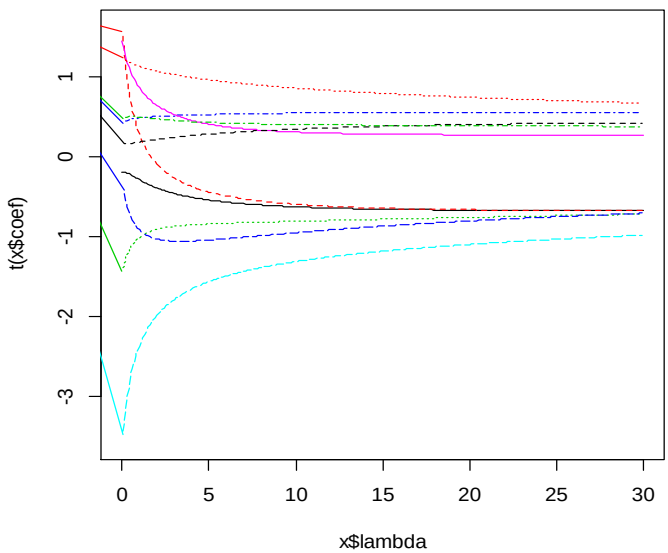
```
> setwd("C:\\Users\\baron\\Documents\\Teach\\615 Regression\\Data")
> HOMES = read.csv("HOME_SALES.csv")
> attach(HOMES)
> reg = lm( SALES_PRICE ~ .-ID-STYLE+as.factor(STYLE), data=HOMES )
> vif(reg)
                     GVIF Df GVIF^(1/(2*Df))
FINISHED_AREA    4.471114  1        2.114501
BEDROOMS         1.689177  1        1.299683
BATHROOMS        3.175930  1        1.782114
GARAGE_SIZE      1.659708  1        1.288296
YEAR_BUILT       1.941969  1        1.393545
LOT_SIZE         1.178829  1        1.085739
AIR_CONDITIONER  1.395114  1        1.181150
POOL             1.057359  1        1.028280
QUALITY          3.817176  2        1.397769
HIGHWAY          1.032433  1        1.016087
as.factor(STYLE) 2.185423  2        1.215861
```

2. Ridge Regression

 # You can run ridge regression with a given chosen lambda or try a whole range of lambdas.

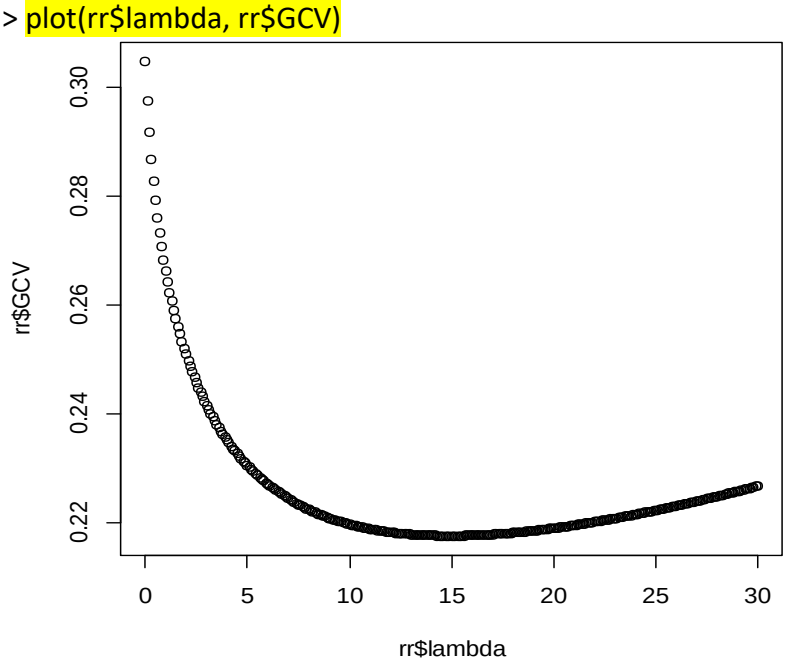
```
> install.packages("MASS")
> library(MASS)
> rr = lm.ridge( mpg ~ ., data=mtcars, lambda=0.5 )
> rr
                       cyl          disp            hp          drat            wt
14.853945860 -0.126513088  0.005661242 -0.016779725  0.886565240 -2.859953631
        qsec            vs            am          gear          carb
 0.607130107  0.341162437  2.396463996  0.693482725 -0.472630120

> rr = lm.ridge( mpg ~ ., data=mtcars, lambda=seq(0,30,0.1) )
> plot(rr)
```



 # These are ridge regression slopes as functions of λ
 # You can also look at the mean-squared prediction error obtained by cross-validation

```
> plot(rr$lambda, rr$GCV)
```



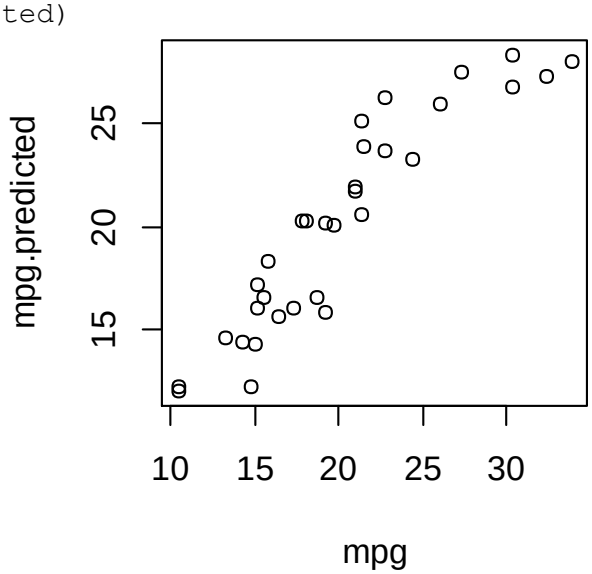
 # Which λ minimize MSPE? Here is how you can find it.

```
> select(rr)
modified HKB estimator is 2.58585
modified L-W estimator is 1.837435
smallest value of GCV  at 14.9

> rr.optimal = lm.ridge( mpg ~ ., data=mtcars, lambda=14.9 )
> coef(rr.optimal)
                    cyl        disp          hp        drat          wt
21.13638336 -0.37299085 -0.00527183 -0.01159978  1.05345707 -1.23062378
       qsec          vs          am        gear        carb
 0.16168503  0.77096188  1.62029355  0.54427196 -0.54685394
```

 # Prediction with ridge regression.
 # lm.ridge does not have a "predict" command…

```
> predict(rr.optimal, data=mtcars)
Error in UseMethod("predict") :
  no applicable method for 'predict' applied to an object of class "ridgelm"
```

 # But we know that ridge regression Is also a linear regression. So, we can compute predicted values "mpg.predicted" simply by the formula X*b, where X = design matrix and b = vector of slopes.
 # Here is an easy way to create matrix X .

```
> reg = lm( mpg ~ ., data=mtcars )
> X = model.matrix(reg)
> X
                    (Intercept) cyl  disp  hp drat    wt  qsec vs am gear carb
Mazda RX4                     1   6 160.0 110 3.90 2.620 16.46  0  1    4    4
Mazda RX4 Wag                 1   6 160.0 110 3.90 2.875 17.02  0  1    4    4
Datsun 710                    1   4 108.0  93 3.85 2.320 18.61  1  1    4    1
Hornet 4 Drive                1   6 258.0 110 3.08 3.215 19.44  1  0    3    1
   < truncated >

> mpg.predicted = X %*% coef(rr.optimal)
> plot(mpg, mpg.predicted)
```